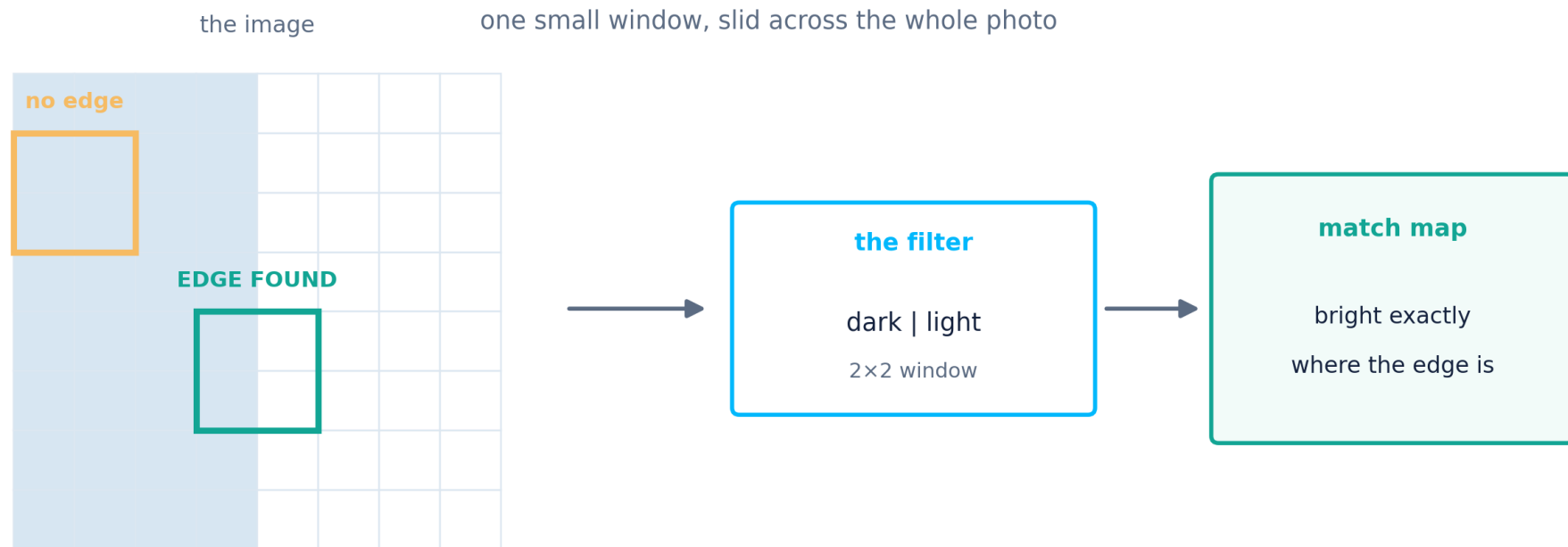


A filter is a tiny sliding window

FINDING A VERTICAL EDGE



The SAME small window slides everywhere — that is why CNNs need so little data

One small window, slid across everything

A CNN does not look at the whole image. It slides a **small window over it.**

- 1 The window is a few pixels wide and looks for one pattern — say a vertical edge.
- 2 It slides across the entire image, marking where that pattern appears.
- 3 Stack many windows and you detect many patterns at once.

REMEMBER

Sliding the same small filter everywhere is why CNNs need so little data per feature.

